

Chapter 9: Differentiation

EXERCISE 9.1 [PAGE 120]

Exercise 9.1 | Q 1.1 | Page 120

Find the derivative of the following function w.r.t. x .
 x^{12}

SOLUTION

$$\text{Let } y = x^{12}$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{d}{dx} x^{12}$$

$$= 12 x^{12-1}$$

$$= 12 x^{11}$$

Exercise 9.1 | Q 1.2 | Page 120

Find the derivative of the following function w.r.t. x .
 x^{-9}

SOLUTION

$$\text{Let } y = x^{-9}$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{d}{dx} x^{-9}$$

$$= -9 x^{-9-1}$$

$$= -9 x^{-10}$$

Find the derivative of the following functions w. r. t. x.

$$x^{\frac{3}{2}}$$

SOLUTION

$$\text{Let } y = x^{\frac{3}{2}}$$

Differentiating w.r.t. x, we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} x^{\frac{3}{2}} \\ &= \frac{3}{2} x^{\frac{3}{2}-1} \\ &= -\frac{3}{2} x^{\frac{1}{2}} \\ &= \frac{3}{2} \sqrt{x}\end{aligned}$$

Find the derivative of the following function w. r. t. x.

$$7x\sqrt{x}$$

SOLUTION

$$\text{Let } y = 7x\sqrt{x}$$

$$= 7x^1 x^{\frac{1}{2}}$$

$$y = 7x^{\frac{3}{2}}$$

Differentiating w.r.t. x, we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} 7x^{\frac{3}{2}} \\ &= 7 \times \frac{3}{2} x^{\frac{3}{2}-1}\end{aligned}$$

$$= \frac{21}{2} x^{\frac{1}{2}}$$

$$= \frac{21}{2} \sqrt{x}$$

Exercise 9.1 | Q 1.5 | Page 120

Find the derivative of the following function w. r. t. x.
 3^5

SOLUTION

$$\text{Let } y = 3^5$$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx} 3^5 = 0 \dots [3^5 \text{ is a constant}]$$

Exercise 9.1 | Q 2.1 | Page 120

Differentiate the following w. r. t. x.
 $x^5 + 3x^4$

SOLUTION

$$\text{Let } y = x^5 + 3x^4$$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx} (x^5 + 3x^4)$$

$$= \frac{d}{dx} x^5 + 3 \frac{d}{dx} x^4$$

$$= 5x^4 + 3(4x^3)$$

$$\frac{dy}{dx} = 5x^4 + 12x^3$$

Differentiate the following w. r. t. x.

$$x\sqrt{x} + \log x - e^x$$

SOLUTION

$$\text{Let } y = x\sqrt{x} + \log x - e^x$$

$$= x^{\frac{3}{2}} + \log x - e^x$$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx} \left(x^{\frac{3}{2}} + \log x - e^x \right)$$

$$= \frac{d}{dx} x^{\frac{3}{2}} + \frac{d}{dx} \log x - \frac{d}{dx} e^x$$

$$= \frac{3}{2} x^{\frac{3}{2}-1} + \frac{1}{x} - e^x$$

$$= \frac{3}{2} x^{\frac{1}{2}} + \frac{1}{x} - e^x$$

$$= \frac{3}{2} \sqrt{x} + \frac{1}{x} - e^x$$

Differentiate the following w. r. t. x.

$$x^{\frac{5}{2}} + 5x^{\frac{7}{5}}$$

SOLUTION

$$\text{Let } y = x^{\frac{5}{2}} + 5x^{\frac{7}{5}}$$

Differentiating w.r.t. x, we get

$$= \frac{dy}{dx} = \frac{d}{dx} \left(x^{\frac{5}{2}} + 5x^{\frac{7}{5}} \right)$$

$$\begin{aligned}
&= \frac{d}{dx} x^{\frac{5}{2}} + 5 \frac{d}{dx} x^{\frac{7}{5}} \\
&= \frac{5}{2} x^{\frac{5}{2}-1} + 5 \frac{7}{5} x^{\frac{7}{5}-1} \\
&= \frac{5}{2} x^{\frac{3}{2}} + 7x^{\frac{2}{5}}
\end{aligned}$$

Exercise 9.1 | Q 2.4 | Page 120

Differentiate the following w. r. t. x.

$$\frac{2}{7}x^{\frac{7}{2}} + \frac{5}{2}x^{\frac{2}{5}}$$

SOLUTION

$$\text{Let } y = \frac{2}{7}x^{\frac{7}{2}} + \frac{5}{2}x^{\frac{2}{5}}$$

Differentiating w.r.t. x, we get

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{2}{7}x^{\frac{7}{2}} + \frac{5}{2}x^{\frac{2}{5}} \right) \\
&= \frac{2}{7} \frac{d}{dx} x^{\frac{7}{2}} + \frac{5}{2} \frac{d}{dx} x^{\frac{2}{5}} \\
&= \frac{2}{7} \times \frac{7}{2} x^{\frac{7}{2}-1} + \frac{5}{2} \times \frac{2}{5} x^{\frac{2}{5}-1} \\
&= x^{\frac{5}{2}} + x^{\frac{-3}{5}}
\end{aligned}$$

Exercise 9.1 | Q 2.5 | Page 120

Differentiate the following w. r. t. x.

$$\sqrt{x}(x^2 + 1)^2$$

SOLUTION

$$\text{Let } y = \sqrt{x}(x^2 + 1)^2$$

$$\therefore y = x^{\frac{1}{2}}(x^4 + 2x^2 + 1)$$

$$y = x^{\frac{9}{2}} + 2x^{\frac{5}{2}} + x^{\frac{1}{2}}$$

Differentiating w.r.t. x, we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left(x^{\frac{9}{2}} + 2x^{\frac{5}{2}} + x^{\frac{1}{2}} \right) \\ &= \frac{d}{dx^{\frac{9}{2}}} + 2 \frac{d}{dx} x^{\frac{5}{2}} + \frac{d}{dx} \sqrt{x} \\ &= \frac{9}{2} x^{\frac{9}{2}-1} + 2 \times \frac{5}{2} x^{\frac{5}{2}-1} + \frac{1}{2\sqrt{x}} \\ &= \frac{9}{2} \frac{x^7}{2} + 5 \frac{x^3}{2} + \frac{1}{2\sqrt{x}} \end{aligned}$$

Exercise 9.1 | Q 3.1 | Page 120

Differentiate the following w. r. t. x

$$x^3 \log x$$

SOLUTION

$$\text{Let } y = (x^3 \log x)$$

Differentiating w.r.t. x, we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} x^3 \log x \\ &= x^3 \frac{d}{dx} (\log x) + (\log x) \frac{d}{dx} (x^3) \\ &= x^3 \times \frac{1}{x} + (\log x)(3x^2) \\ &= x^2 + 3x^2 \log x \end{aligned}$$

Differentiate the following w. r. t. x

$$x^{\frac{5}{2}} e^x$$

SOLUTION

$$\text{Let } y = x^{\frac{5}{2}} e^x$$

Differentiating w.r.t. x, we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(x^{\frac{5}{2}} e^x \right) \\ &= x^{\frac{5}{2}} \frac{d}{dx} (e^x) + e^x \left(\frac{5}{2} x^{\frac{3}{2}} \right) \\ &= x^{\frac{5}{2}} (e^x) + e^x \left(\frac{5}{2} x^{\frac{3}{2}} \right) \\ &= e^x \left(x^{\frac{5}{2}} + \frac{5}{2} x^{\frac{3}{2}} \right)\end{aligned}$$

Differentiate the following w. r. t. x

$$e^x \log x$$

SOLUTION

$$\text{Let } y = e^x \log x$$

Differentiating w.r.t. x, we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (e^x \log x) \\ &= e^x \frac{d}{dx} (\log x) + (\log x) \frac{d}{dx} (e^x)\end{aligned}$$

$$= e^x \left(\frac{1}{x} \right) + (\log x)(e^x)$$

$$= e^x \left(\frac{1}{x} + \log x \right)$$

Exercise 9.1 | Q 3.4 | Page 120

Differentiate the following w. r. t. x
 $x^3 \cdot 3^x$

SOLUTION

$$\text{Let } y = x^3 \cdot 3^x$$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx} (x^3 \cdot 3^x)$$

$$= x^3 \frac{d}{dx} (3^x) + 3^x \frac{d}{dx} (x^3)$$

$$= (x^3)(3^x \log 3) + 3^x(3x^2)$$

$$= x^2 \cdot 3^x (x \log 3 + 3)$$

Exercise 9.1 | Q 4.1 | Page 120

Find the derivative of the following w. r. t. x

$$\frac{x^2 + a^2}{x^2 - a^2}$$

SOLUTION

$$\text{Let } y = \frac{x^2 + a^2}{x^2 - a^2}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{x^2 + a^2}{x^2 - a^2} \right) \\&= \frac{(x^2 - a^2) \frac{d}{dx} (x^2 + a^2) - (x^2 + a^2) \frac{d}{dx} (x^2 - a^2)}{(x^2 - a^2)^2} \\&= \frac{(x^2 - a^2) \left(\frac{d}{dx} x^2 + \frac{d}{dx} a^2 \right) - (x^2 + a^2) \left(\frac{d}{dx} x^2 - \frac{d}{dx} a^2 \right)}{(x^2 - a^2)^2} \\&= \frac{(x^2 - a^2)(2x + 0) - (x^2 + a^2)(2x - 0)}{(x^2 - a^2)^2} \\&= \frac{2x(x^2 - a^2) - 2x(x^2 + a^2)}{(x^2 - a^2)^2} \\&= \frac{2x(x^2 - a^2 - x^2 - a^2)}{(x^2 - a^2)^2} \\&= \frac{2x(-2a^2)}{(x^2 - a^2)^2} \\&= \frac{-4xa^2}{(x^2 - a^2)^2}\end{aligned}$$

Exercise 9.1 | Q 4.2 | Page 120

Find the derivative of the following w. r. t. x .

$$\frac{3x^2 + 5}{2x^2 - 4}$$

SOLUTION

$$\text{Let } y = \frac{3x^2 + 5}{2x^2 - 4}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{3x^2 + 5}{2x^2 - 4} \right) \\&= \frac{(2x^2 - 4) \frac{d}{dx} (3x^2 + 5) - (3x^2 + 5) \frac{d}{dx} (2x^2 - 4)}{(2x^2 - 4)^2} \\&= \frac{(2x^2 - 4)(6x + 0) - (3x^2 + 5)(4x - 0)}{(2x^2 - 4)^2} \\&= \frac{6x(2x^2 - 4) - 4x(3x^2 + 5)}{(2x^2 - 4)^2} \\&= \frac{2x[3(2x^2 - 4) - 2(3x^2 + 5)]}{(2x^2 - 4)^2} \\&= \frac{2x(6x^2 - 12 - 6x^2 - 10)}{(2x^2 - 4)^2} \\&= \frac{2x(-22)}{(2x^2 - 4)^2} \\&= \frac{-44x}{(2x^2 - 4)^2}\end{aligned}$$

Exercise 9.1 | Q 4.3 | Page 120

Find the derivative of the following w. r. t. x

$$\frac{\log x}{x^3 - 5}$$

SOLUTION

$$\text{Let } y = \frac{\log x}{x^3 - 5}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{\log x}{x^3 - 5} \right) \\&= \frac{(x^3 - 5) \frac{d}{dx} (\log x) - (\log x) \frac{d}{dx} (x^3 - 5)}{(x^3 - 5)^2} \\&= \frac{(x^3 - 5) \left(\frac{1}{x} \right) - \log x \left(\frac{d}{dx} (x^3) - \frac{d}{dx} (5) \right)}{(x^3 - 5)^2} \\&= \frac{(x^3 - 5) \frac{1}{x} - \log x (3x^2 - 0)}{(x^3 - 5)^2} \\&= \frac{(x^3 - 5) \frac{1}{x} - 3x^2 \log x}{(x^3 - 5)^2}\end{aligned}$$

Exercise 9.1 | Q 4.4 | Page 120

Find the derivative of the following w. r. t. x .

$$\frac{3e^x - 2}{3e^x + 2}$$

SOLUTION

$$\text{Let } y = \frac{3e^x - 2}{3e^x + 2}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{3e^x - 2}{3e^x + 2} \right) \\&= \frac{(3e^x + 2) \frac{d}{dx} (3e^x - 2) - (3e^x - 2) \frac{d}{dx} (3e^x + 2)}{(3e^x + 2)^2}\end{aligned}$$

$$\begin{aligned}
&= \frac{(3e^x)\left(\frac{d}{dx}(3e^x) - \frac{d}{dx}(2)\right) - (3e^x - 2)\left(\frac{d}{dx}(3e^x) + \frac{d}{dx}(2)\right)}{(3e^x + 2)^2} \\
&= \frac{(3e^x + 2)(3e^x - 0) - (3e^x - 2)(3e^x + 0)}{(3e^x + 2)^2} \\
&= \frac{3e^x(3e^x + 2) - 3e^x(3e^x - 2)}{(3e^x + 2)^2} \\
&= \frac{3e^x(3e^x + 2 - 3e^x + 2)}{(3e^x + 2)^2} \\
&= \frac{3e^x(4)}{(3e^x + 2)^2} \\
&= \frac{12e^x}{(3e^x + 2)^2}
\end{aligned}$$

Exercise 9.1 | Q 4.5 | Page 120

Find the derivative of the following w. r. t.x.

$$\frac{xe^x}{x + e^x}$$

SOLUTION

$$\text{Let } y = \frac{xe^x}{x + e^x}$$

Differentiating w.r.t. x, we get

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{xe^x}{x + e^x} \right) \\
&= \frac{(x + e^x) \frac{d}{dx}(xe^x) - (xe^x) \frac{d}{dx}(x + e^x)}{(x + e^x)^2} \\
&= \frac{(x + e^x) \left[x \frac{d}{dx}(e^x) + e^x \frac{d}{dx}(x) \right] - xe^x \left(\frac{d}{dx}(x) + \frac{d}{dx}(e^x) \right)}{(x + e^x)^2}
\end{aligned}$$

$$\begin{aligned}
&= \frac{(x + e^x)[xe^x + e^x(1)] - xe^x(1 + e^x)}{(x + e^x)^2} \\
&= \frac{(x + e^x)(xe^x + e^x) - xe^x(1 + e^x)}{(x + e^x)^2} \\
&= \frac{(x + e^x)e^x(x + 1) - xe^x(1 + e^x)}{(x + e^x)^2} \\
&= \frac{e^x[(x + e^x)(x + 1) - x(1 + e^x)]}{(x + e^x)^2}
\end{aligned}$$

Exercise 9.1 | Q 5.1 | Page 120

Find the derivative of the following function by the first principle.
 $3x^2 + 4$

SOLUTION

$$\text{Let } f(x) = 3x^2 + 4$$

$$\therefore f(x + h) = 3(x + h)^2 + 4$$

$$= 3(x^2 + 2xh + h^2) + 4$$

$$= 3x^2 + 6xh + 3h^2 + 4$$

By first principle, we get

$$\begin{aligned}
f'(x) &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{(3x^2 + 6xh + 3h^2 + 4) - (3x^2 + 4)}{h} \\
&= \lim_{h \rightarrow 0} \frac{3h^2 + 6xh}{h} \\
&= \lim_{h \rightarrow 0} \frac{h(3h + 6x)}{h}
\end{aligned}$$

$$= \lim_{h \rightarrow 0} (6x + 3h) \quad \dots [\because h \rightarrow 0, \therefore h \neq 0]$$

$$= 6x + 3(0)$$

$$= 6x$$

Exercise 9.1 | Q 5.2 | Page 120

Find the derivative of the following function by the first principle.

$$x\sqrt{x}.$$

SOLUTION

$$\text{Let } f(x) = x\sqrt{x} = x^{\frac{3}{2}}$$

$$\therefore f(x+h) = (x+h)^{\frac{3}{2}}$$

By first principle, we get

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^{\frac{3}{2}} - x^{\frac{3}{2}}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\left[(x+h)^{\frac{3}{2}} - x^{\frac{3}{2}}\right] \left[(x+h)^{\frac{3}{2}} + x^{\frac{3}{2}}\right]}{h \left[(x+h)^{\frac{3}{2}} + x^{\frac{3}{2}}\right]} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h \left[(x+h)^{\frac{3}{2}} + x^{\frac{3}{2}}\right]} \\ &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h \left[(x+h)^{\frac{3}{2}} + x^{\frac{3}{2}}\right]} \end{aligned}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2)}{h \left[(x+h)^{\frac{3}{2}} + x^{\frac{3}{2}} \right]} \\
&= \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2)}{h \left[(x+h)^{\frac{3}{2}} + x^{\frac{3}{2}} \right]} \\
&= \lim_{h \rightarrow 0} \frac{3x^2 + 3xh + h^2}{(x+h)^{\frac{3}{2}} + x^{\frac{3}{2}}} \quad \dots [\because h \rightarrow 0, \therefore h \neq 0] \\
&= \frac{3x^2 + 3 \times x0 + 0^2}{(x+0)^{\frac{3}{2}} + x^{\frac{3}{2}}} \\
&= \frac{3x^2}{2x^{\frac{3}{2}}} \\
&= \frac{3}{2} x^{\frac{1}{2}} \\
&= \frac{3}{2} \sqrt{x}
\end{aligned}$$

Exercise 9.1 | Q 5.3 | Page 120

Find the derivative of the following functions by the first principle.

$$\frac{1}{2x+3}$$

SOLUTION

$$\text{Let } f(x) = \frac{1}{2x+3}$$

$$\therefore f(x+h) = \frac{1}{2(x+h)+3} = \frac{1}{2x+2h+3}$$

By first principle, we get

$$\begin{aligned}
f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{\left(\frac{1}{2x+2h+3}\right) - \left(\frac{1}{2x+3}\right)}{h} \\
&= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{2x+3 - 2x-2h-3}{(2x+2h+3)(2x+3)} \right] \\
&= \lim_{h \rightarrow 0} \frac{1}{h} \frac{-2h}{(2x+2h+3)(2x+3)} \\
&= \lim_{h \rightarrow 0} \frac{-2}{(2x+2h+3)(2x+3)} \quad \dots [\because h \rightarrow 0, \therefore h \neq 0] \\
&= \frac{-2}{(2x+2 \times 0+3)(2x+3)} \\
&= \frac{-2}{(2x+3)^2}
\end{aligned}$$

Exercise 9.1 | Q 5.4 | Page 120

Find the derivative of the following function by the first principle.

$$\frac{x-1}{2x+7}$$

SOLUTION

$$\text{Let } f(x) = \frac{x-1}{2x+7}$$

$$\therefore f(x+h) = \frac{x+h-1}{2(x+h)+7} = \frac{x+h-1}{2x+2h+7}$$

By first principle, we get

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{\frac{x+h+1}{2x+2h+7} - \frac{x-1}{2x+7}}{h} \\
&= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{(x+h-1)(2x+7) - (x-1)(2x+2h+7)}{(2x+2h+7)(2x+7)} \right] \\
&= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{(2x^2 + 2xh - 2x + 7x + 7h - 7 - 2x^2 - 2xh - 7x + 2x + 2h + 7)}{(2x+2h+7)(2x+7)} \right] \\
&= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{9h}{(2x+2h+7)(2x+7)} \right] \\
&= \frac{9}{(2x+2 \times 0+7)(2x+7)} \\
&= \frac{9}{(2x+7)^2}
\end{aligned}$$

EXERCISE 9.2 [PAGES 122 - 123]

Exercise 9.2 | Q 1.1 | Page 122

Differentiate the following function w.r.t.x.

$$\frac{x}{x+1}$$

SOLUTION

$$\text{Let } y = \frac{x}{x+1}$$

Differentiating w.r.t. x, we get

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{x}{x+1} \right) \\
&= \frac{(x+1) \frac{d}{dx}(x) - x \frac{d}{dx}(x+1)}{(x+1)^2}
\end{aligned}$$

$$\begin{aligned}
 &= \frac{(x+1)(1) - x(1+0)}{(x+1)^2} \\
 &= \frac{x+1-x}{(x+1)^2} \\
 &= \frac{1}{(x+1)^2}
 \end{aligned}$$

Exercise 9.2 | Q 1.2 | Page 122

Differentiate the following function w.r.t.x

$$\frac{x^2 + 1}{x}$$

SOLUTION

$$\text{Let } y = \frac{x^2 + 1}{x}$$

Differentiating w.r.t. x, we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{x^2 + 1}{x} \right) \\
 &= \frac{x \frac{d}{dx} (x^2 + 1) - (x^2 + 1) \frac{d}{dx} (x)}{x^2} \\
 &= \frac{x(2x + 0) - (x^2 + 1)(1)}{x^2} \\
 &= \frac{2x^2 - x^2 - 1}{x^2} \\
 \frac{dy}{dx} &= \frac{x^2 - 1}{x^2}
 \end{aligned}$$

Differentiate the following function w.r.t.x.

$$\frac{1}{e^x + 1}$$

SOLUTION

$$\text{Let } y = \frac{1}{e^x + 1}$$

Differentiating w.r.t. x, we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{1}{e^x + 1} \right) \\&= \frac{(e^x + 1) \frac{d}{dx}(1) - (1) \frac{d}{dx}(e^x + 1)}{(e^x + 1)^2} \\&= \frac{(e^x + 1)(0) - (1)(e^x + 0)}{(e^x + 1)^2} \\&= \frac{e^x + 1 - e^x}{(e^x + 1)^2} \\&= \frac{1}{(e^x + 1)^2}\end{aligned}$$

Differentiate the following function w.r.t.x

$$\frac{e^x}{e^x + 1}$$

SOLUTION

$$y = \frac{e^x}{e^x + 1}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{e^x}{e^x + 1} \right) \\&= \frac{(e^x + 1) \frac{d}{dx}(e^x) - \frac{d}{dx}(e^x + 1)}{(e^x + 1)^2} \\&= \frac{(e^x + 1)e^x - e^x(e^x + 0)}{(e^x + 1)^2} \\&= \frac{e^x(e^x + 1 - e^x)}{(e^x + 1)^2} \\&= \frac{e^x}{(e^x + 1)^2}\end{aligned}$$

Exercise 9.2 | Q 1.5 | Page 122

Differentiate the following function w.r.t. x

$$\frac{x}{\log x}$$

SOLUTION

$$\text{Let } y = \frac{x}{\log x}$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{x}{\log x} \right)$$

$$\begin{aligned}
 &= \frac{\log x \frac{d}{dx}(x) - x \frac{d}{dx}(\log x)}{(\log x)^2} \\
 &= \frac{\log x(1) - x\left(\frac{1}{x}\right)}{(\log x)^2} \\
 &= \frac{\log x - 1}{(\log x)^2}
 \end{aligned}$$

Exercise 9.2 | Q 1.6 | Page 122

Differentiate the following function w.r.t.x.

$$\frac{2^x}{\log x}$$

SOLUTION

$$\text{Let } y = \frac{2^x}{\log x}$$

Differentiating w.r.t. x, we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{2^x}{\log x} \right) \\
 &= \frac{\log x \frac{d}{dx}(2^x) - 2^x \frac{d}{dx}(\log x)}{(\log x)^2} \\
 &= \frac{\log x(2^x \log 2) - 2^x\left(\frac{1}{x}\right)}{(\log x)^2} \\
 &= \frac{(2^x \log x \cdot \log 2) \left(-\frac{1}{x}\right)}{(\log x)^2}
 \end{aligned}$$

Differentiate the following function w.r.t.x

$$\frac{(2e^x - 1)}{(2e^x + 1)}$$

SOLUTION

$$\text{Let } y = \frac{2e^x - 1}{2e^x + 1}$$

Differentiating w.r.t. x, we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{2e^x - 1}{2e^x + 1} \right) \\ &= \frac{(2e^x + 1) \frac{d}{dx} (2e^x - 1) - (2e^x - 1) \frac{d}{dx} (2e^x + 1)}{(2e^x + 1)^2} \\ &= \frac{(2e^x + 1)(2e^x) - (2e^x - 1)(2e^x)}{(2e^x + 1)^2} \\ &= \frac{2e^x(2e^x + 1 - 2e^x + 1)}{(2e^x + 1)^2} \\ &= \frac{2e^x(2)}{(2e^x + 1)^2} \\ &= \frac{4e^x}{(2e^x + 1)^2} \end{aligned}$$

Differentiate the following function w.r.t.x

$$\frac{(x + 1)(x - 1)}{(e^x + 1)}$$

SOLUTION

$$\text{Let } y = \frac{(x+1)(x-1)}{(e^x+1)}$$

$$\therefore y = \frac{x^2-1}{(e^x+1)}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{x^2-1}{e^x+1} \right) \\&= \frac{(e^x+1) \frac{d}{dx} (x^2-1) - (x^2-1) \frac{d}{dx} (e^x+1)}{(e^x+1)^2} \\&= \frac{(x^2+1)(2x) - (x^2-1)(e^x+0)}{(e^x+1)^2} \\&= \frac{2xe^x + 2x - x^2e^x + e^x}{(e^x+1)^2} \\&= \frac{2xe^x + e^x - x^2e^x + 2x}{(e^x+1)^2} \\&= \frac{e^x(2x+1-x^2) + 2x}{(e^x+1)^2}\end{aligned}$$

Exercise 9.2 | Q 2.01 | Page 122

Solve the following example:

The demand D for a price P is given as $D = 27/P$, find the rate of change of demand when price is 3.

SOLUTION

$$\text{Demand, } D = \frac{27}{P}$$

$$\text{Rate of change of demand} = \frac{dD}{dP}$$

$$= \frac{d}{dP} \left(\frac{27}{p} \right)$$

$$= 27 \frac{d}{dP} P \left(\frac{1}{p} \right)$$

$$= 27 \frac{d}{dP} \left(\frac{1}{P} \right)$$

$$= 27 \frac{d}{dP} (P^{-1})$$

$$= 27((-1)P^{-2})$$

$$= 27 \left(\frac{-1}{p^2} \right) = \frac{-27}{p^2}$$

When price $P = 3$,

Rate of change of demand,

$$\left(\frac{dD}{dP} \right)_{P=3} = \frac{-27}{(3)^2} = -3$$

$\therefore$ When price is 3, Rate of change of demand is -3.

Exercise 9.2 | Q 2.02 | Page 122

Solve the following example:

If for a commodity; the price-demand relation is given as $D = \frac{P+5}{P-1}$. Find the marginal demand when price is 2.

SOLUTION

$$\text{Given, } D = \frac{P + 5}{P - 1}$$

$$\begin{aligned} \text{Marginal demand} &= \frac{dD}{dP} = \frac{d}{dP} \left(\frac{P + 5}{P - 1} \right) \\ &= \frac{(P - 1) \frac{d}{dP} (P + 5) - (P + 5) \frac{d}{dP} (P - 1)}{(P - 1)^2} \\ &= \frac{(P - 1)(1 + 0) - (P + 5)(1 - 0)}{(P - 1)^2} \\ &= \frac{P - 1 - P - 5}{(P - 1)^2} \\ &= \frac{-6}{(P - 1)^2} \end{aligned}$$

When $P = 2$,

$$\begin{aligned} \text{Marginal demand, } \left(\frac{dD}{dP} \right)_{P=2} &= \frac{-6}{(2 - 1)^2} = -6 \end{aligned}$$

$\therefore$ When price is 2, marginal demand is -6.

Exercise 9.2 | Q 2.03 | Page 122

Solve the following example:

The demand function of a commodity is given as $P = 20 + D - D^2$. Find the rate at which price is changing when demand is 3.

SOLUTION

Given, $P = 20 + D - D^2$

Rate of change of price = $\frac{dP}{dD}$

$$= \frac{d}{dD} (20 + D - D^2)$$

$$= 0 + 1 - 2D$$

$$= 1 - 2D$$

Rate of change of price at $D = 3$ is

$$\left(\frac{dP}{dD} \right)_{D=3}$$

$$= 1 - 2(3) = -5$$

∴ Price is changing at a rate of -5 when demand is 3.

Exercise 9.2 | Q 2.04 | Page 122

Solve the following example:

If the total cost function is given by; $C = 5x^3 + 2x^2 + 7$; find the average cost and the marginal cost when $x = 4$.

SOLUTION

Total cost function, $C = 5x^3 + 2x^2 + 7$

$$\text{Average cost} = \frac{C}{x}$$

$$= \frac{5x^3 + 2x^2 + 7}{x}$$

$$= 5x^2 + 2x + \frac{7}{x}$$

When $x = 4$,

$$\text{Average cost} = 5(4)^2 + 2(4) + \frac{7}{4}$$

$$= 80 + 8 + \frac{7}{4}$$

$$= \frac{320 + 32 + 7}{4}$$

$$= \frac{359}{4}$$

$$\text{Marginal cost} = \frac{dC}{dx}$$

$$= \frac{d}{dx} (5x^3 + 2x^2 + 7)$$

$$= 5 \frac{d}{dx} (x^3) + 2 \frac{d}{dx} (x^2) + \frac{d}{dx} (7)$$

$$= 5(3x^2) + 2(2x) + 0$$

$$= 15x^2 + 4x$$

$$\text{When } x = 4, \text{ Marginal cost} = \left(\frac{dC}{dx} \right)_{x=4}$$

$$= 15(4)^2 + 4(4)$$

$$= 240 + 16$$

$$= 256$$

$\therefore$ the average cost and marginal cost at $x = 4$ are $\frac{359}{4}$ and 256 respectively.

Exercise 9.2 | Q 2.05 | Page 122

Solve the following example:

The total cost function of producing n notebooks is given by $C = 1500 - 75n + 2n^2 + \frac{n^3}{5}$.

Find the marginal cost at $n = 10$.

SOLUTION

Total cost function,

$$C = 1500 - 75n + 2n^2 + \frac{n^3}{5}$$

$$\text{Marginal Cost} = \frac{dC}{dn}$$

$$= \frac{d}{dn} \left(1500 - 75n + 2n^2 + \frac{n^3}{5} \right)$$

$$= \frac{d}{dn} (1500) - 75 \frac{d}{dn} (n) + 2 \frac{d}{dn} (n^2) + \frac{1}{5} \frac{d}{dn} (n^3)$$

$$= 0 - 75(1) + 2(2n) + \frac{1}{5} (3n^2)$$

$$= -75 + 4n + \frac{3n^2}{5}$$

When $n = 10$,

Marginal cost

$$\begin{aligned}
&= \left(\frac{dC}{dn} \right)_{n=10} = -75 + 4(10) + \frac{3}{5}(10)^2 \\
&= -75 + 40 + 60 \\
&= 25
\end{aligned}$$

Exercise 9.2 | Q 2.06 | Page 123

Solve the following example:

The total cost of 't' toy cars is given by $C=5(2^t) + 17$. Find the marginal cost and average cost at $t=3$.

SOLUTION

Total cost of 't' toy cars, $C = 5(2^t) + 17$

$$\begin{aligned}
\text{Marginal Cost} &= \frac{dC}{dt} \\
&= \frac{d}{dt} [5(2^t) + 17] \\
&= 5 \frac{d}{dt} (2^t) + \frac{d}{dt} (17) \\
&= 5(2^t \cdot \log 2) + 0
\end{aligned}$$

$$= 5(2^t \cdot \log 2)$$

When $t = 3$,

$$\text{Marginal cost} = \left(\frac{dC}{dt} \right)_{t=3}$$

$$= 5(2^3 \cdot \log 2) = 40 \log 2$$

$$\text{Average cost} = \frac{C}{t} = \frac{5(2)^t + 17}{t}$$

$$= \frac{40 + 17}{3} = 19$$

∴ at $t = 3$, Marginal cost is 40 log 2 and Average cost is 19.

Exercise 9.2 | Q 2.07 | Page 123

Solve the following example:

If for a commodity; the demand function is given by, $D = \sqrt{75 - 3P}$. find the marginal demand function when $P = 5$

SOLUTION

Demand function, $D = \sqrt{75 - 3P}$

Now, Marginal demand = $\frac{dD}{dP}$

$$= \frac{d}{dP} (\sqrt{75 - 3P})$$

$$= \frac{1}{2\sqrt{75 - 3P}} \cdot \frac{d}{dP} (75 - 3P)$$

$$= \frac{1}{2\sqrt{75 - 3P}} \cdot (0 - 3 \times 1)$$

$$= \frac{-3}{2\sqrt{75 - 3P}}$$

When $P = 5$,

$$\text{Marginal demand} = \left(\frac{dD}{dP} \right)_{P=5}$$

$$= \frac{-3}{2\sqrt{75 - 3(5)}}$$

$$= \frac{-3}{2\sqrt{60}}$$

$$= \frac{-3}{4\sqrt{15}}$$

$$\therefore \text{Marginal demand} = \frac{-3}{4\sqrt{15}} \text{ at } P = 5.$$

Exercise 9.2 | Q 2.08 | Page 123

Solve the following example:

The total cost of producing x units is given by $C=10e^{2x}$, find its marginal cost and average cost when $x = 2$

SOLUTION

$$\text{Total cost, } C = 10e^{2x}$$

$$\text{Marginal cost} = \frac{dC}{dx}$$

$$= \frac{d}{dx} (10e^2x) = 10 \frac{d}{dx} (e^2x)$$

$$= 10 \cdot e^2x \cdot \frac{d}{dx} (2x) = 10 \cdot e^2x \cdot 2(1)$$

$$= 20e^{2x}$$

When $x = 2$,

$$\text{Marginal cost} = \left(\frac{dC}{dx} \right)_{x=2}$$

$$= 20e^4$$

$$\text{Average cost} = \frac{C}{x}$$

$$= \frac{10e^2x}{x}$$

$$\text{When } x = 2 \text{ average cost} = \frac{10e^4}{2} = 5e^4$$

$\therefore$ When $x = 2$, marginal cost is $20e^4$ and average cost is $5e^4$.

Exercise 9.2 | Q 2.09 | Page 123

Solve the following example:

The demand function is given as $P = 175 + 9D + 25D^2$. Find the revenue, average revenue, and marginal revenue when demand is 10.

SOLUTION

$$\text{Given, } P = 175 + 9D + 25D^2$$

$$\text{Total revenue, } R = P \cdot D$$

$$= (175 + 9D + 25D^2)D$$

$$= 175D + 9D^2 + 25D^3$$

$$\text{Average revenue} = P = 175 + 9D + 25D^2$$

$$\text{Marginal revenue} = \frac{dR}{dD}$$

$$= \frac{d}{dD} (175D + 9D^2 + 25D^3)$$

$$= 175 \frac{d}{dD} (D) + 9 \frac{d}{dD} D(D^2) + 25 \frac{d}{dD} (D^3)$$

$$= 175(1) + 9(2D) + 25(3D^2)$$

$$= 175 + 18D + 75D^2$$

$$\text{When } D = 10,$$

$$\text{Total revenue} = 175(10) + 9(10)^2 + 25(10)^3$$

$$= 1750 + 900 + 25000 = 27650$$

$$\text{Average revenue} = 175 + 9(10) + 25(10)^2$$

$$= 175 + 90 + 2500 = 2765$$

$$\text{Marginal revenue} = 175 + 18(10) + 75(10)^2$$

$$= 175 + 180 + 7500 = 7855$$

∴ When Demand = 10,

Total revenue = 27650,

Average revenue = 2765

Marginal revenue = 7855.

Exercise 9.2 | Q 2.1 | Page 123

Solve the following example:

The supply S for a commodity at price P is given by $S = P^2 + 9P - 2$. Find the marginal supply when price is 7.

SOLUTION

$$\text{Given, } S = P^2 + 9P - 2$$

$$\text{Marginal supply} = \frac{dS}{dP}$$

$$= \frac{d}{dP} (P^2 + 9P - 2)$$

$$= \frac{d}{dP} (P^2) + 9 \frac{d}{dP} (P) - \frac{d}{dP} (2)$$

$$= 2P + 9(1) - 0$$

$$= 2P + 9$$

When $P = 7$,

$$\text{Marginal supply} = \left(\frac{dS}{dP} \right)_{P=7}$$

$$= 2(7) + 9$$

$$= 14 + 9 = 23$$

∴ Marginal supply is 23, at $P = 7$.

Exercise 9.2 | Q 2.11 | Page 123

Solve the following example:

The cost of producing x articles is given by $C = x^2 + 15x + 81$. Find the average cost and marginal cost functions. Find marginal cost when $x = 10$. Find x for which the marginal cost equals the average cost.

SOLUTION

Given, cost $C = x^2 + 15x + 81$

$$\text{Average cost} = \frac{C}{x} = \frac{x^2 + 15x + 81}{x}$$

$$= x + 15 + \frac{81}{x}$$

$$\text{and Marginal cost} = \frac{dC}{dx}$$

$$= \frac{d}{dx} (x^2 + 15x + 81)$$

$$= \frac{d}{dx} (x^2) + 15 \frac{d}{dx} (x) + \frac{d}{dx} (81)$$

$$= 2x + 15(1) + 0 = 2x + 15$$

When $x = 10$,

$$\text{Marginal cost} = \left(\frac{dC}{dx} \right)_{x=10}$$

$$= 2(10) + 15 = 35$$

If marginal cost = average cost, then

$$2x + 15 = x + 15 + \frac{81}{x}$$

$$\therefore x = \frac{81}{x}$$

$$\therefore x^2 = 81$$

$$\therefore x = 9 \quad \dots [\because x > 0]$$

MISCELLANEOUS EXERCISE 9 [PAGES 123 - 124]

Miscellaneous Exercise 9 | Q 1.1 | Page 123

Differentiate the following function .w.r.t.x
 x^5

SOLUTION

$$\text{Let } y = x^5$$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx} x^5 = 5x^4$$

Miscellaneous Exercise 9 | Q 1.2 | Page 123

Differentiate the following function w.r.t.x
 x^{-2}

SOLUTION

$$\text{Let } y = x^{-2}$$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{d}{dx} (x^{-2}) = -2x^{-3} = \frac{-2}{x^3}$$

Miscellaneous Exercise 9 | Q 1.3 | Page 123

Differentiate the following functions w.r.t.x.

$$\sqrt{x}$$

SOLUTION

$$\text{Let } y = \sqrt{x}$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$$

Miscellaneous Exercise 9 | Q 1.4 | Page 123

Differentiate the following function w.r.t. x

$$x\sqrt{x}$$

SOLUTION

$$\text{Let } y = x\sqrt{x}$$

$$\therefore y = x^{\frac{3}{2}}$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{d}{dx} x^{\frac{3}{2}} = \frac{3}{2} x^{\frac{1}{2}}$$

Miscellaneous Exercise 9 | Q 1.5 | Page 123

Differentiate the following functions w.r.t. x .

$$\frac{1}{\sqrt{x}}$$

SOLUTION

$$\text{Let } y = \frac{1}{\sqrt{x}}$$

$$\therefore y = x^{-\frac{1}{2}}$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{-1}{2} x^{-\frac{3}{2}} = \frac{-1}{2x^{\frac{3}{2}}}$$

Miscellaneous Exercise 9 | Q 1.6 | Page 123

Differentiate the following functions. w.r.t. x
 7^x

SOLUTION

$$\text{Let } y = 7^x$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{d}{dx} 7^x = 7^x \log 7$$

Miscellaneous Exercise 9 | Q 2.01 | Page 123

Find $\frac{dy}{dx}$ if

$$y = x^2 + \frac{1}{x^2}$$

SOLUTION

$$y = x^2 + \frac{1}{x^2}$$

$$\therefore y = x^2 + x^{-2}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^2 + x^{-2}) \\ &= \frac{d}{dx}(x^2) + \frac{d}{dx}(x^{-2}) \\ &= 2x - 2x^{-3} \\ &= 2x - \frac{2}{x^3}\end{aligned}$$

Miscellaneous Exercise 9 | Q 2.02 | Page 123

Find $\frac{dy}{dx}$ if
 $y = (\sqrt{x} + 1)^2$

SOLUTION

$$y = (\sqrt{x} + 1)^2$$

$$\therefore y = x + 2\sqrt{x} + 1$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x + 2\sqrt{x} + 1) \\ &= \frac{d}{dx}(x) + 2\frac{d}{dx}(\sqrt{x}) + \frac{d}{dx}(1)\end{aligned}$$

$$\begin{aligned}
 &= 1 + 2\left(\frac{1}{2\sqrt{x}}\right) + 0 \\
 &= \frac{dy}{dx} = 1 + \frac{1}{\sqrt{x}}
 \end{aligned}$$

Miscellaneous Exercise 9 | Q 2.03 | Page 123

Find $\frac{dy}{dx}$ if

$$y = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2$$

SOLUTION

$$y = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2$$

$$\therefore y = x + 2 + \frac{1}{x}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} \left(x + 2 + \frac{1}{x} \right) \\
 &= \frac{d}{dx}(x) + \frac{d}{dx}(2) + \frac{d}{dx} \left(\frac{1}{x} \right) \\
 &= 1 + 0 + \frac{d}{dx}(x^{-1}) \\
 &= 1 + (-1)x^{-2} \\
 &= 1 - \frac{1}{x^2}
 \end{aligned}$$

Find $\frac{dy}{dx}$ if

$$y = x^3 - 2x^2 + \sqrt{x} + 1$$

SOLUTION

$$y = x^3 - 2x^2 + \sqrt{x} + 1$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (x^3 - 2x^2 + \sqrt{x} + 1) \\ &= \frac{d}{dx} (x^3) - 2 \frac{d}{dx} (x^2) + \frac{d}{dx} (\sqrt{x}) + \frac{d}{dx} (1) \\ &= 3x^2 - 2(2x) + \frac{d}{dx} (x^{\frac{1}{2}}) + 0 \\ &= 3x^2 - 4x + \frac{1}{2} x^{\frac{1}{2}-1} \\ &= 3x^2 - 4x + \frac{1}{2} x^{-\frac{1}{2}} \\ \frac{dy}{dx} &= 3x^2 - 4x + \frac{1}{2\sqrt{x}}\end{aligned}$$

Find $\frac{dy}{dx}$ if

$$y = x^2 + 2^x - 1$$

SOLUTION

$$y = x^2 + 2^x - 1$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^2 + 2^x - 1) \\ &= \frac{d}{dx}(x^2) + \frac{d}{dx}(2^x) - \frac{d}{dx}(1) \\ &= 2x + 2^x \log 2 - 0 \\ &= 2x + 2^x \log 2\end{aligned}$$

Miscellaneous Exercise 9 | Q 2.06 | Page 123

Find $\frac{dy}{dx}$ if

$$y = (1 - x)(2 - x)$$

SOLUTION

$$\begin{aligned}y &= (1 - x)(2 - x) \\ &= 2 - 3x + x^2\end{aligned}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(2 - 3x + x^2) \\ &= \frac{d}{dx}(2) - 3\frac{d}{dx}(x) + \frac{d}{dx}(x^2) \\ &= 0 - 3(1) + 2x \\ &= -3 + 2x\end{aligned}$$

Find $\frac{dy}{dx}$ if

$$y = \frac{1+x}{2+x}$$

SOLUTION

$$y = \frac{1+x}{2+x}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{1+x}{2+x} \right) \\ &= \frac{(2+x) \frac{d}{dx}(1+x) - (1+x) \frac{d}{dx}(2+x)}{(2+x)^2} \\ &= \frac{(2+x)(0+1) - (1+x)(0+1)}{(2+x)^2} \\ \frac{dy}{dx} &= \frac{(2+x) - (1+x)}{(2+x)^2} \\ &= \frac{2+x-1-x}{(2+x)^2} \\ &= \frac{1}{(2+x)^2}\end{aligned}$$

Find $\frac{dy}{dx}$ if

$$y = \frac{(\log x + 1)}{x}$$

SOLUTION

$$y = \frac{(\log x + 1)}{x}$$

Differentiating w.r.t. x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left[\frac{\log x + 1}{x} \right] \\&= \frac{x \frac{d}{dx} (\log x + 1) - (\log x + 1) \frac{d}{dx} (x)}{x^2} \\&= \frac{x \left(\frac{1}{x} + 0 \right) - (\log x + 1)(1)}{x^2} \\&= \frac{1 - \log x - 1}{x^2} \\&= \frac{-\log x}{x^2}\end{aligned}$$

Miscellaneous Exercise 9 | Q 2.09 | Page 123

Find $\frac{dy}{dx}$ if

$$y = \frac{e^x}{\log x}$$

SOLUTION

$$y = \frac{e^x}{\log x}$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{e^x}{\log x} \right)$$

$$\begin{aligned}
&= \frac{(\log x) \frac{d}{dx}(e^x) - (e^x) \frac{d}{dx}(\log x)}{(\log x)^2} \\
&= \frac{(\log x)e^x - e^x\left(\frac{1}{x}\right)}{(\log x)^2} \\
&= \frac{e^x\left(\log x - \frac{1}{x}\right)}{(\log x)^2}
\end{aligned}$$

Miscellaneous Exercise 9 | Q 2.1 | Page 123

Find $\frac{dy}{dx}$ if

$$y = x \log x (x^2 + 1)$$

SOLUTION

$$y = x \log x (x^2 + 1)$$

Differentiating w.r.t. x , we get

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx}(x)(\log x)(x^2 + 1) \\
&= (x)(\log x) \frac{d}{dx}(x^2 + 1) - (x^2 + 1) \frac{d}{dx}((x)(\log x)) \\
&= (x \log x)(2x + 0) + (x^2 + 1) \left[x \frac{d}{dx}(\log x) + (\log x) \frac{d}{dx}(x) \right] \\
&= 2x^2 \log x + (x^2 + 1) \left[x \times \frac{1}{x} + (\log x)(1) \right] \\
&= 2x^2 \log x + (x^2 + 1)(1 + \log x) \\
&= 2x^2 \log x + (x^2 + 1) + (x^2 + 1) \log x
\end{aligned}$$

Miscellaneous Exercise 9 | Q 3.01 | Page 124

Solve the following.

The relation between price (P) and demand (D) of a cup of Tea is given by $D = 12/P$. Find the rate at which the demand changes when the price is Rs. 2/- Interpret the result.

SOLUTION

$$\text{Demand, } D = \frac{12}{P}$$

$$\text{Rate of change of demand} = \frac{dD}{dP}$$

$$= \frac{d}{dP} \left(\frac{12}{P} \right)$$

$$= 12 \frac{d}{dP} (P^{-1}) = 12((-1)P^{-2})$$

$$= 12 \left(\frac{-1}{P^2} \right) = \frac{-12}{P^2}$$

When price $P = 2$,

$$\text{Rate of change of demand, } \left(\frac{dD}{dP} \right)_{P=2} = \frac{-12}{(2)^2} = -3$$

$\therefore$ When price is 2, Rate of change of demand is -3

Here, rate of change of demand is negative

$\therefore$ demand would fall when the price becomes ₹ 2.

Miscellaneous Exercise 9 | Q 3.02 | Page 124

Solve the following.

The demand (D) of biscuits at price P is given by $D = 64/P^3$, find the marginal demand when price is Rs. 4/-.

SOLUTION

$$\text{Given demand } D = \frac{64}{P^3}$$

$$\text{Now, marginal demand} = \frac{dD}{dP}$$

$$= \frac{d}{dP} \left(\frac{64}{p^3} \right)$$

$$= 64 \frac{d}{dP} (p^{-3})$$

$$= 64 (-3) P^{-4}$$

$$= \frac{-192}{p^4}$$

When $P = 4$

$$\text{Marginal demand} = \left(\frac{dD}{dP} \right)_{p=4}$$

$$= \frac{-192}{(4)^4}$$

$$= \frac{-192}{256}$$

$$= \frac{-3}{4}$$

Miscellaneous Exercise 9 | Q 3.03 | Page 124

Solve the following:

The supply S of electric bulbs at price P is given by $S = 2P^3 + 5$. Find the marginal supply when the price is Rs. 5/- Interpret the result.

SOLUTION

Given, supply $S = 2p^3 + 5$

Now, marginal supply $= \frac{dS}{dp}$

$$= \frac{d}{dp} (2p^3 + 5)$$

$$= 2 \frac{d}{dp} (p^3) + \frac{d}{dp} (5)$$

$$= 2(3p^2) + 0$$

$$= 6p^2$$

∴ When $p = 5$

$$\text{Marginal supply} = \left(\frac{dS}{dp} \right)_{p=5}$$

$$= 6(5)^2 = 150$$

Here, the rate of change of supply with respect to the price is positive which indicates that the supply increases.

Miscellaneous Exercise 9 | Q 3.04 | Page 124

Solve the following:

The marginal cost of producing x items is given by $C = x^2 + 4x + 4$. Find the average cost and the marginal cost. What is the marginal cost when $x = 7$.

SOLUTION

Total cost, $C = x^2 + 4x + 4$

$$\text{Now, Average cost} = \frac{c}{x} = \frac{x^2 + 4x + 4}{x}$$

$$= x + 4 + \frac{4}{x}$$

$$\text{and Marginal cost} = \frac{dc}{dx} \frac{d}{dx} (x^2 + 4x + 4)$$

$$= \frac{d}{dx}(x^2) + 4 \frac{d}{dx}(x) + \frac{d}{dx}(4)$$

$$= 2x + 4(1) + 0$$

$$= 2x + 4$$

∴ When $x = 7$,

$$\text{Marginal cost} = \left(\frac{dC}{dx} \right)_{x=7}$$

$$= 2(7) + 4$$

$$= 14 + 4$$

$$= 18$$

Miscellaneous Exercise 9 | Q 3.05 | Page 124

Solve the following:

The Demand D for a price P is given as $D = 27/P$, Find the rate of change of demand when the price is Rs. 3/-.

SOLUTION

$$\text{Demand, } D = \frac{27}{P}$$

$$\text{Rate of change of demand} = \frac{dD}{dP}$$

$$= \frac{d}{dP} \left(\frac{27}{P} \right)$$

$$= 27 \frac{d}{dP} \left(\frac{1}{P} \right)$$

$$= 27 \frac{d}{dP} (P^{-1})$$

$$= 27((-1)P^{-2})$$

$$= 27 \left(\frac{-1}{p^2} \right) = \frac{-27}{p^2}$$

When price $P = 3$,

Rate of change of demand,

$$\left(\frac{dD}{dP} \right)_{P=3} = \frac{-27}{(3)^2} = -3$$

$\therefore$ When price is 3, Rate of change of demand is -3.

Miscellaneous Exercise 9 | Q 3.06 | Page 124

Solve the following.

If for a commodity; the price demand relation is given by $D = \left(\frac{P+5}{P-1} \right)$. Find the marginal demand when price is Rs. 2/-.

SOLUTION

$$\text{Given, } D = \left(\frac{P+5}{P-1} \right)$$

$$\text{Marginal demand} = \left(\frac{dD}{dP} \right) = \frac{d}{dP} \left(\frac{P+5}{P-1} \right)$$

$$= \frac{(P-1) \frac{d}{dP}(P+5) - (P+5) \frac{d}{dP}(P-1)}{(P-1)^2}$$

$$= \frac{(P-1)(1+0) - (P+5)(1-0)}{(P-1)^2}$$

$$= \frac{P-1-P-5}{(P-1)^2}$$

$$= \frac{-6}{(P-1)^2}$$

When $P = 2$,

$$\begin{aligned}\text{Marginal demand, } \left(\frac{dP}{dP} \right)_{P=2} \\&= \frac{-6}{(2-1)^2} \\&= -6\end{aligned}$$

$\therefore$ When price is 2, marginal demand is -6.

Miscellaneous Exercise 9 | Q 3.07 | Page 124

Solve the following.

The price function P of a commodity is given as $P = 20 + D - D^2$ where D is demand. Find the rate at which price (P) is changing when demand $D = 3$.

SOLUTION

$$\text{Given, } P = 20 + D - D^2$$

$$\text{Rate of change of price} = \frac{dP}{dD}$$

$$= \frac{d}{dD} (20 + D - D^2)$$

$$= 0 + 1 - 2D$$

$$= 1 - 2D$$

Rate of change of price at $D = 3$ is

$$\left(\frac{dP}{dD} \right)_{D=3} = 1 - 2(3) = -5$$

$\therefore$ Price is changing at a rate of -5 when demand is 3.

Miscellaneous Exercise 9 | Q 3.08 | Page 124

Solve the following.

If the total cost function is given by $C = 5x^3 + 2x^2 + 1$; Find the average cost and the marginal cost when $x = 4$.

SOLUTION

$$\text{Total cost function } C = 5x^3 + 2x^2 + 1$$

$$\text{Average cost} = \frac{C}{x}$$

$$= \frac{5x^3 + 2x^2 + 1}{x}$$

$$= 5x^2 + 2x + \frac{1}{x}$$

$$\text{When } x = 4, \text{ Average cost} = 5(4)^2 + 2(4) + \frac{1}{4}$$

$$= 80 + 8 + \frac{1}{4}$$

$$= \frac{320 + 32 + 1}{4}$$

$$= \frac{353}{4}$$

$$\text{Marginal cost} = \frac{dC}{dx}$$

$$= \frac{d}{dx} (5x^3 + 2x^2 + 1)$$

$$= 5 \frac{d}{dx} (x^3) + 2 \frac{d}{dx} (x^2) + \frac{d}{dx} (1)$$

$$= 5(3x^2) + 2(2x) + 0$$

$$= 15x^2 + 4x$$

$$\text{When } x = 4, \text{ marginal cost} = \left(\frac{dC}{dx} \right)_{x=4}$$

$$= 15(4)^2 + 4(4)$$

$$= 240 + 16$$

$$= 256$$

∴ The average cost and marginal cost at $x = 4$ are $\frac{353}{4}$ and 256 respectively.

Miscellaneous Exercise 9 | Q 3.09 | Page 124

Solve the following.

The supply S for a commodity at price P is given by $S = P^2 + 9P - 2$. Find the marginal supply when price Rs. 7/-.

SOLUTION

$$\text{Given, } S = P^2 + 9P - 2$$

$$\text{Marginal supply} = \frac{dS}{dP}$$

$$= \frac{d}{dP} (P^2 + 9P - 2)$$

$$= \frac{d}{dP} (P^2) + 9 \frac{d}{dP} (P) - \frac{d}{dP} (2)$$

$$= 2P + 9(1) - 0$$

$$= 2P + 9$$

$$\text{When } P = 7,$$

$$\text{Marginal supply} = \left(\frac{dS}{dP} \right)_{P=7}$$

$$= 2(7) + 9$$

$$= 14 + 9 = 23$$

∴ Marginal supply is 23, at $P = 7$.

Miscellaneous Exercise 9 | Q 3.1 | Page 124

Solve the following.

The cost of producing x articles is given by $C = x^2 + 15x + 81$. Find the average cost and marginal cost functions. Find the marginal cost when $x = 10$. Find x for which the marginal cost equals the average cost.

SOLUTION

Given, cost $C = x^2 + 15x + 81$

$$\text{Average cost} = \frac{C}{x} = \frac{x^2 + 15x + 81}{x}$$

$$= x + 15 + \frac{81}{x}$$

$$\text{and Marginal cost} = \frac{dC}{dx}$$

$$= \frac{d}{dx}(x^2 + 15x + 81)$$

$$= \frac{d}{dx}(x^2) + 15\frac{d}{dx}(x) + \frac{d}{dx}(81)$$

$$= 2x + 15(1) + 0 = 2x + 15$$

When $x = 10$,

$$\text{Marginal cost} = \left(\frac{dC}{dx} \right)_{x=10}$$

$$= 2(10) + 15 = 35$$

If marginal cost = average cost, then

$$2x + 15 = x + 15 + \frac{81}{x}$$

$$\therefore x = \frac{81}{x}$$

$$\therefore x^2 = 81$$

$$\therefore x = 9 \quad \dots [\because x > 0]$$